Equilibrium Simulation: Mathematical Framework

Magnani & Munro (2020) with Treatment-Specific Payoffs

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Reading guide. Sections marked with ★ extend the Magnani & Munro (2020) framework. All other material replicates or directly follows their model.

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1 Model

We adapt the dynamic market runs model of Magnani & Munro (2020, hereafter M&M) to our experimental design. The game consists of $N = 4$ risk-averse investors, each holding one unit of an asset. Time is discrete, and the game terminates each period with probability λ. Investors observe public signals about a binary state $z \in \{G, B\}$ and decide whether to sell irreversibly.

★ Our innovation is to compare two payoff treatments—*Random* and *Average*—that differ in how simultaneous sellers split the available prices. M&M analyze only the Random treatment; we extend the model to derive equilibrium predictions for both treatments and characterize how the treatment affects equilibrium behavior.

1.1 Notation

Symbol	Definition
$N = 4$	Number of investors
$n \in \{1, 2, 3, 4\}$	Number of investors still holding
$\pi \in (0, 1)$	Posterior probability that state is Good, $\Pr(z = G)$
$\sigma(n, \pi)$	Symmetric mixing probability: each holder sells with prob σ
$V(n, \pi)$	Value for a holder at start of period (before signal)
$W(n, \pi')$	Value for a holder after signal observation (at decision point)
$\lambda = 0.125$	Per-period termination probability
$v = 20$	Asset return in good state
$p_n = 2n$	Price when n holders remain: $p_1 = 2,\ p_2 = 4,\ p_3 = 6,\ p_4 = 8$
$\mu_B = 0.675$	$\Pr(\text{bad signal} \mid z = B)$
$\mu_G = 0.325$	$\Pr(\text{bad signal} \mid z = G)$
α	CRRA risk aversion parameter
$u(x) = \frac{x^{1-\alpha}}{1-\alpha}$	CRRA utility ($u(x) = x$ if $\alpha = 0$; $u(x) = \ln x$ if $\alpha = 1$)

1.2 Information structure

Agents start with common prior $\pi_0 = 0.5$. Each period, a public signal $y \in \{g, b\}$ is drawn with accuracy $\mu_B + \mu_G = 1$. We adopt the M&M normalization so that good and bad signals are exact Bayesian inverses. The posterior updates are:

$$\pi_g(\pi) = \frac{\pi(1 - \mu_G)}{\pi(1 - \mu_G) + (1 - \pi)\mu_G} \qquad \text{(after good signal)} \tag{1}$$

$$\pi_b(\pi) = \frac{\pi(1 - \mu_B)}{\pi(1 - \mu_B) + (1 - \pi)\mu_B} \qquad \text{(after bad signal)} \tag{2}$$

The unconditional signal probabilities given belief π are:

$$\Pr(y = g \mid \pi) = (1 - \mu_G)\pi + \mu_G(1 - \pi) \tag{3}$$

$$\Pr(y = b \mid \pi) = (1 - \mu_B)\pi + \mu_B(1 - \pi) \tag{4}$$

1.3 Belief grid

Since $\mu_B + \mu_G = 1$, the likelihood ratio update from a bad signal is the exact multiplicative inverse of a good signal:

$$\frac{\Pr(y = b \mid B)}{\Pr(y = b \mid G)} = \frac{\mu_B}{\mu_G} = \frac{0.675}{0.325} \approx 2.077 = \left(\frac{\mu_G}{\mu_B}\right)^{-1} = \left(\frac{\Pr(y = g \mid B)}{\Pr(y = g \mid G)}\right)^{-1}$$

Consequently, after k bad and j good signals from $\pi_0 = 0.5$, the posterior depends only on the net bad count $d = k - j$:

$$\pi(d) = \frac{1}{1 + \left(\frac{\mu_B}{\mu_G}\right)^d} \tag{5}$$

The reachable belief grid consists of $2T_{\max}+1$ values, computed by applying $d \in \{-T_{\max}, \ldots, T_{\max}\}$. We set $T_{\max} = 20$, yielding a 41-point grid spanning from $\pi(-20) \approx 4.5 \times 10^{-7}$ to $\pi(20) \approx 1 - 4.5 \times 10^{-7}$. The grid is non-uniform in π but uniform in log-odds space.

Remark 1 (Grid sufficiency)**.** *Robustness checks in Section 11 confirm that both sigma values and equilibrium thresholds are fully converged by $T_{\max} = 20$: doubling to $T_{\max} = 40, 80, 160$ produces changes below 10^{-3}.*

2 Payoff structure

2.1 Voluntary selling

When k investors sell simultaneously from n remaining holders, the prices consumed are $\{p_n, p_{n-1}, \ldots, p_{n-k+1}\}$. ★ The payoff depends on the treatment:

Random treatment (M&M baseline). Each seller is randomly assigned one of the k positions with equal probability, receiving the corresponding price:

$$\rho_{\text{random}}(n, k) = \frac{1}{k} \sum_{j=0}^{k-1} u(p_{n-j}) \tag{6}$$

This is the expected utility of a uniform lottery over the k available prices.

★ **Average treatment.** Each seller receives the average price:

$$\rho_{\text{average}}(n, k) = u\left(\frac{1}{k} \sum_{j=0}^{k-1} p_{n-j}\right) \tag{7}$$

This is the utility of the certain average price.

For $k = 1$, both treatments are identical: $\rho(n, 1) = u(p_n)$. For $k > 1$ and $\alpha > 0$, Jensen's inequality gives $\rho_{\text{average}} > \rho_{\text{random}}$ since u is strictly concave. The difference is:

$$\rho_{\text{average}}(n, k) - \rho_{\text{random}}(n, k) = u(\mathbb{E}[p]) - \mathbb{E}[u(p)] > 0 \tag{8}$$

This "risk premium" in selling payoffs is one of two channels through which the treatment affects equilibrium behavior (Section 5).

2.2 Forced liquidation

If the game terminates in the bad state with m holders remaining, all are forced to sell at prices $\{p_1, \ldots, p_m\}$:

Random: $L_{\text{random}}(m) = \frac{1}{m}\sum_{i=1}^{m} u(p_i)$ ★ **Average:** $L_{\text{average}}(m) = u\left(\frac{1}{m}\sum_{i=1}^{m} p_i\right)$

Again, $L_{\text{average}}(m) \geq L_{\text{random}}(m)$ by Jensen's inequality for $\alpha > 0$. This "insurance" in liquidation payoffs is the second channel (Section 5).

3 Bellman equations

3.1 Pre-signal value

At the start of a period, before the signal is observed:

$$V(n, \pi) = \Pr(y{=}g \mid \pi) \cdot W\big(n, \pi_g(\pi)\big) + \Pr(y{=}b \mid \pi) \cdot W\big(n, \pi_b(\pi)\big) \tag{9}$$

where signal probabilities are given by Eqs. (3)–(4).

3.2 Post-signal value

After observing the signal and updating to π', each holder sells with probability $\sigma = \sigma(n, \pi')$:

$$W(n, \pi') = \sigma \cdot U_{\text{sell}}(n, \sigma) + (1 - \sigma) \cdot U_{\text{hold}}(n, \pi', \sigma) \tag{10}$$

At an interior equilibrium ($\sigma \in (0, 1)$), the indifference condition $U_{\text{sell}} = U_{\text{hold}}$ implies $W = U_{\text{sell}} = U_{\text{hold}}$, so the weighting is immaterial. At corner solutions ($\sigma = 0$ or $\sigma = 1$), only one term is active.

3.3 Expected utility of selling

The investor sells along with j of the other $n-1$ holders (each selling independently with probability σ):

$$U_{\text{sell}}(n, \sigma) = \sum_{j=0}^{n-1} \binom{n-1}{j} \sigma^j (1 - \sigma)^{n-1-j} \cdot \rho(n,\, j + 1) \tag{11}$$

Note that U_{sell} does not depend on π'—the selling payoff is realized immediately and does not depend on the state. The total number of sellers is $j + 1$ (the focal seller plus the j others).

3.4 Expected utility of holding

$$U_{\text{hold}}(n, \pi', \sigma) = \sum_{j=0}^{n-1} \binom{n-1}{j} \sigma^j (1 - \sigma)^{n-1-j} \cdot H(n{-}j,\, \pi') \tag{12}$$

where $m = n - j$ investors remain after j others sell (the focal holder stays). Note the summation runs over $j = 0, \ldots, n - 1$ because at most $n - 1$ *others* can sell.

3.5 Continuation value

After j others sell, $m = n - j$ holders remain. With probability λ the game terminates (good state yields v, bad state forces liquidation); with probability $1 - \lambda$ the game continues:

$$H(m, \pi') = \lambda\big[\pi' \cdot u(v) + (1 - \pi') \cdot L(m)\big] + (1 - \lambda) \cdot V(m, \pi') \tag{13}$$

This creates the circular dependency that makes the model non-trivial: V depends on W, W depends on U_{hold}, U_{hold} depends on H, and H depends on V.

4 Equilibrium

We solve for *symmetric Markov-perfect equilibria*: all investors use the same strategy $\sigma_i(n, \pi) = \sigma(n, \pi)$.

4.1 Equilibrium conditions

For each state (n, π'):

1. If $U_{\text{sell}}(n, 0) \leq U_{\text{hold}}(n, \pi', 0)$: $\sigma = 0$ (holding dominates)
2. If $U_{\text{sell}}(n, 1) \geq U_{\text{hold}}(n, \pi', 1)$: $\sigma = 1$ (selling dominates)
3. Otherwise: $\sigma \in (0, 1)$ solves $U_{\text{sell}}(n, \sigma) = U_{\text{hold}}(n, \pi', \sigma)$ (indifference)

4.2 Boundary condition

For $n = 1$ (last holder): $\sigma(1, \pi) = 0$ for all π. Selling yields $u(p_1) = u(2)$, while holding preserves the option value $H(1, \pi') = \lambda[\pi' u(v) + (1 - \pi')u(p_1)] + (1 - \lambda)V(1, \pi')$. Since $u(v) = u(20) \gg u(2)$, holding always dominates when there is no strategic interaction. The last holder has no coordination risk and simply waits for the game to end.

4.3 Uniqueness

Proposition 1. *For each state (n, π') with $n \geq 2$, the equilibrium mixing probability σ^* is unique.*

Argument. Define $\Delta(\sigma) \equiv U_{\text{sell}}(n, \sigma) - U_{\text{hold}}(n, \pi', \sigma)$. The equilibrium condition requires $\Delta(\sigma^*) = 0$ (or a corner solution). We argue Δ is strictly decreasing in σ:

U_{sell} is strictly decreasing in σ. The selling payoff is $\rho(n, j+1)$ where $j \sim \text{Binomial}(n-1, \sigma)$ is the number of others who sell simultaneously. Since $\rho(n, k)$ is decreasing in k (more simultaneous sellers means each receives a weakly lower price), increasing σ shifts the distribution of j upward in the sense of first-order stochastic dominance, reducing U_{sell}.

U_{hold} is also decreasing in σ, but more slowly. When σ increases, the holder transitions to a state with fewer remaining holders ($m = n - j$ lower in expectation). The continuation value $H(m, \pi')$ is increasing in m because:

- The good-state terminal payoff $u(v)$ does not depend on m.
- The forced-liquidation value $L(m)$ is increasing in m since the price schedule $p_m = 2m$ is increasing, so fewer holders means worse liquidation outcomes.

- The continuation value $V(m, \pi')$ is increasing in m: lower m means lower available selling prices and lower forced-liquidation payoffs in all future periods.

Thus higher σ (which shifts weight toward lower m) reduces U_{hold}.

Both utilities decrease, but U_{sell} decreases faster. Uniqueness requires that U_{sell} is more sensitive to simultaneous competition than U_{hold}. Intuitively, a seller is *directly* affected by the number of simultaneous sellers (splitting the current prices), while a holder is only *indirectly* affected through future continuation values. We verify numerically that $\Delta(\sigma)$ is strictly decreasing at every grid point (n, π') across all parameter configurations $(\alpha, \text{treatment})$, confirming at most one root in $(0, 1)$.

This yields a unique equilibrium σ^* at each state—a *strategic substitutes* property that rules out the self-reinforcing panics arising in bank run models with strategic complements. □

Remark 2 (Scope of uniqueness)**.** *The argument above establishes uniqueness of σ^* at each individual state (n, π'), given the value function V. Combined with the contraction mapping property of value iteration (verified by convergence from both low and high initial conditions to the same fixed point), this implies a unique symmetric Markov-perfect equilibrium. However, asymmetric equilibria where different investors use different strategy functions could in principle exist. The strategic-substitutes property makes this unlikely, but we restrict attention to symmetric strategies throughout.*

4.4 Threshold structure

For each $n \geq 2$, define $\pi^*(n)$ as the belief where $\Delta(0) = U_{\text{sell}}(n, 0) - U_{\text{hold}}(n, \pi^*, 0) = 0$. Then:

- For $\pi > \pi^*(n)$: $\sigma(n, \pi) = 0$ (pure hold)
- For $\pi \leq \pi^*(n)$: $\sigma(n, \pi) > 0$ (mixing or pure sell)
- $\sigma(n, \pi)$ is weakly decreasing in π (more pessimistic beliefs $\Rightarrow$ more selling)
- $\pi^*(n)$ is increasing in n (more holders $\Rightarrow$ selling starts at higher beliefs, reflecting greater coordination risk)

We compute $\pi^*(n)$ by root-finding (Brent's method) on the function $\pi \mapsto U_{\text{sell}}(n, 0) - U_{\text{hold}}(n, \pi, 0)$, yielding a continuous threshold that does not depend on grid placement.

5 ★ Treatment effects

★ This section extends M&M by deriving how the Average treatment changes equilibrium behavior relative to the Random treatment.

5.1 Risk-neutral equivalence

Under risk neutrality ($\alpha = 0$), $u(x) = x$ is linear, so Jensen's inequality has no bite: $\rho_{\text{random}} = \rho_{\text{average}}$ and $L_{\text{random}} = L_{\text{average}}$. The equilibrium is identical across treatments. This provides a useful check: any difference between treatments at $\alpha = 0$ would indicate a bug.

5.2 Two competing channels

Under risk aversion ($\alpha > 0$), the Average treatment changes payoffs through two channels that work in opposite directions:

1. **Selling channel.** $\rho_{\text{average}}(n, k) > \rho_{\text{random}}(n, k)$ for $k > 1$ by Jensen's inequality (Eq. 8). Receiving a certain average price is preferred to a lottery over heterogeneous prices under concave utility. This makes selling more attractive in the Average treatment, pushing σ upward and π^* upward.

2. **Holding channel.** $L_{\text{average}}(m) > L_{\text{random}}(m)$ for $m > 1$ by the same Jensen's inequality. Forced liquidation under the Average treatment is less painful because holders receive the average of the forced prices rather than a random assignment. This increases the continuation value $H(m, \pi')$ in Eq. (13), making holding more attractive. This pushes σ downward and π^* downward.

5.3 Net effect: holding channel dominates

Quantitatively, the holding channel dominates: equilibrium thresholds $\pi^*(n)$ are lower under the Average treatment, meaning the selling region is *narrower*. Investors are willing to hold longer because the worst-case outcome (forced liquidation in the bad state) is less painful.

However, the two channels create a subtle interaction. Conditional on the same belief π', the mixing probability σ can be higher under Average (the selling-channel improvement dominates locally at beliefs near the threshold). But the threshold itself shifts, so the overall selling probability—integrating over the belief distribution generated by the signal process—is lower under Average.

The magnitude of the treatment effect is increasing in α:

- At $\alpha = 0$: no treatment effect (Jensen's inequality inactive)
- At $\alpha = 0.5$: moderate threshold differences (a few percentage points in π^*)
- At $\alpha = 0.9$: larger threshold differences

6 Solution algorithm

The equilibrium has a circular structure: V depends on σ through W, and σ depends on V through U_{hold}. We resolve this via value iteration, which repeatedly alternates between updating strategies (given the current value function) and updating the value function (given the current strategies).

6.1 Initialization

Construct the 41-point belief grid (Section 1.3). Store $V(n, \pi)$ and $\sigma(n, \pi)$ as arrays of dimension 4×41 (one entry per (n, π) pair). Initialize $V^{(0)}(n, \pi) = 0$ for all (n, π).

6.2 Iteration

Each iteration t proceeds in two passes:

Pass 1: Update σ and compute W. For each grid point (n, π'):

1. Compute $U_{\text{sell}}(n, 0)$ using Eq. (11). This does not depend on V since selling payoffs are immediate.

2. Compute $U_{\text{hold}}(n, \pi', 0)$ using Eq. (12) with $\sigma = 0$. This *does* depend on $V^{(t)}$ through the continuation value $H(m, \pi')$ in Eq. (13). Since $\sigma = 0$, nobody else sells, so $m = n$ and $U_{\text{hold}} = H(n, \pi')$.

3. **Find $\sigma^{(t+1)}$.** Define $\Delta(\sigma) \equiv U_{\text{sell}}(n, \sigma) - U_{\text{hold}}(n, \pi', \sigma)$, where U_{hold} uses $V^{(t)}$ in the continuation value. Then:

$$\sigma^{(t+1)} = \begin{cases} 0 & \text{if } \Delta(0) \leq 0 \\ 1 & \text{if } \Delta(0) > 0 \text{ and } \Delta(1) \geq 0 \\ \sigma^* \text{ s.t. } \Delta(\sigma^*) = 0 & \text{if } \Delta(0) > 0 \text{ and } \Delta(1) < 0 \end{cases}$$

   Interior roots are found via Brent's method with tolerance 10^{-12}.

4. Compute $W^{(t+1)}(n, \pi') = \sigma \cdot U_{\text{sell}}(n, \sigma) + (1 - \sigma) \cdot U_{\text{hold}}(n, \pi', \sigma)$.

Pass 2: Update V. For each grid point (n, π):

1. Compute the updated beliefs $\pi_g(\pi)$ and $\pi_b(\pi)$ via Eqs. (1)–(2).

2. Look up $W^{(t+1)}(n, \pi_g)$ and $W^{(t+1)}(n, \pi_b)$ by linearly interpolating the W array at the updated beliefs.

3. Set $V^{(t+1)}(n, \pi) = \Pr(y{=}g \mid \pi) \cdot W(n, \pi_g) + \Pr(y{=}b \mid \pi) \cdot W(n, \pi_b)$ per Eq. (9).

6.3 Convergence

Repeat until $\max_{n,\pi} |V^{(t+1)}(n, \pi) - V^{(t)}(n, \pi)| < 10^{-8}$. The output is the converged strategy table $\sigma(n, \pi)$ for all 4×41 state pairs.

Convergence is verified by two additional checks:

- **Initialization independence.** The solver supports both "low" ($V^{(0)} = 0$) and "high" ($V^{(0)} = u(v)/(1 - \alpha)$) initialization. Both converge to the same fixed point, providing a bracketing check on uniqueness.

- **Equilibrium conditions.** At every grid point of the converged solution, we verify that $U_{\text{sell}} = U_{\text{hold}}$ (within 10^{-6}) at interior sigma, and $U_{\text{sell}} \leq U_{\text{hold}}$ wherever $\sigma = 0$.

7 ★ Alternative solution algorithms

★ To verify that our numerical results are not artifacts of the value iteration algorithm, we implement and compare four alternative solution methods. All five approaches converge to the same equilibrium (Section 10).

7.1 Policy iteration (exact V solve)

Given a current strategy $\sigma(n, \pi')$, we solve for V exactly by reformulating the Bellman equation as a linear system. Define the *signal transition matrix* T with entries:

$$T_{ij} = \Pr(\pi_j \mid \pi_i) = \Pr(y{=}g \mid \pi_i) \cdot \mathbf{1}[\pi_j = \pi_g(\pi_i)] + \Pr(y{=}b \mid \pi_i) \cdot \mathbf{1}[\pi_j = \pi_b(\pi_i)] \tag{14}$$

where the indicator is replaced by linear interpolation weights when $\pi_g(\pi_i)$ or $\pi_b(\pi_i)$ falls between grid points. The matrix T is row-stochastic ($\sum_j T_{ij} = 1$), non-negative, and sparse (at most 4 non-zero entries per row).

For fixed σ, the post-signal value $W(n, \pi'_j)$ decomposes into a constant part (selling payoff, terminal values) and a part linear in $V(n, \cdot)$ (continuation via H). This gives:

$$V(n, \cdot) = T \cdot \big[c(n, \cdot) + C(n) \cdot V(n, \cdot)\big] \tag{15}$$

where c is a vector of constants and C is a coefficient matrix. Rearranging:

$$(I - T \cdot C) \cdot V(n, \cdot) = T \cdot c(n, \cdot)$$

which is a standard linear system solved via LU decomposition.

Policy iteration alternates between (1) updating σ at all grid points given V, and (2) solving the linear system for V given σ. This replaces the iterative V-update of value iteration with an exact solve, eliminating any concern about insufficient convergence.

7.2 Howard improvement

A hybrid approach: perform one σ-update step, then run K value iteration sweeps (without updating σ) before the next σ-update. We test $K = 10$ and $K = 50$. This interpolates between value iteration ($K = 1$) and policy iteration ($K = \infty$).

7.3 Discount convention variant

We test an alternative placement of the termination probability. Instead of embedding λ in the continuation value H (Eq. 13), we define:

$$V_{\text{disc}}(n, \pi) = \lambda\big[\pi \cdot u(v) + (1 - \pi) \cdot L(n)\big] + (1 - \lambda) \cdot \big[\Pr(g|\pi) \cdot W(n, \pi_g) + \Pr(b|\pi) \cdot W(n, \pi_b)\big]$$

with $H(m, \pi') = V(m, \pi')$ (no λ in the continuation). These formulations are algebraically equivalent when both converge, but could differ during iteration if the fixed-point path differs.

7.4 W-at-indifference variant

At interior equilibrium ($0 < \sigma < 1$), the indifference condition gives $U_{\text{sell}} = U_{\text{hold}} = W$. We test setting $W = U_{\text{sell}}$ directly (rather than the mixture $\sigma \cdot U_{\text{sell}} + (1 - \sigma) \cdot U_{\text{hold}}$) to check whether the weighting formula matters for convergence.

8 Simulation engine

The equilibrium solution gives a strategy table $\sigma(n, \pi)$—the probability of selling at each state. To produce predictions comparable to experimental data, we need to know when sales actually occur. This requires simulation because the *timing* of sales depends on the stochastic signal process: beliefs random-walk through the grid, and a sale happens only when (a) the belief enters the selling region *and* (b) the coin flip with probability σ comes up "sell." Neither the threshold π^* nor the mixing probability σ alone tells us the average belief at sale.

8.1 One simulated game

For a given $(\alpha, \text{treatment})$ pair with converged strategy table $\sigma(n, \pi)$:

1. **Setup.** Set $n = 4$ holders, belief $\pi = 0.5$. Draw the true state $z \in \{G, B\}$ with equal probability ($\Pr(G) = 0.5$, matching the prior). Initialize seller count $s = 0$.
2. **Each period** (repeat until $n = 1$ or game terminates):
   (a) *Signal.* Draw a signal from the true state: if $z = B$, the signal is bad with probability $\mu_B = 0.675$; if $z = G$, with probability $\mu_G = 0.325$. Update π via Bayesian rule (Eqs. (1)–(2)).
   (b) *Selling decisions.* Look up $\sigma(n, \pi)$ from the strategy table (linearly interpolating if π falls between grid points). Draw $k \sim \text{Binomial}(n, \sigma)$: the number of holders who sell this period.
   (c) *Record sales.* If $k > 0$: for each of the k sellers, record the current belief π as the belief at seller position $s + 1, s + 2, \ldots, s + k$. Update $s \leftarrow s + k$ and $n \leftarrow n - k$.
   (d) *Termination.* With probability $\lambda = 0.125$, the game ends. If the game terminates, stop—any remaining holders did not sell voluntarily.

Note that multiple investors can sell in the same period (simultaneous sales), and they all receive the same belief π for their seller position. Games where the market terminates before any voluntary sale contribute no observations. A safety cap of 200 periods prevents infinite loops; given $\lambda = 0.125$, the probability of reaching 200 periods is $(1 - \lambda)^{200} \approx 2.6 \times 10^{-12}$.

8.2 ★ Aggregation and common random numbers

★ We run 10,000 games per $(\alpha, \text{treatment})$ pair. For each seller position $k \in \{1, 2, 3\}$, we compute the average belief at the time of sale:

$$\bar{\pi}_k = \frac{1}{|\mathcal{G}_k|} \sum_{g \in \mathcal{G}_k} \pi_k^{(g)}$$

where $\mathcal{G}_k$ is the set of games in which a kth sale happened and $\pi_k^{(g)}$ is the belief at that sale. The 4th investor never sells in equilibrium ($\sigma(1, \pi) = 0$ for all π), so we report three seller positions.

We report $\bar{\pi}_k = \Pr(z = G)$ directly, converting to $\Pr(z = B) = 1 - \pi$ only when comparing against M&M Table 2 (which uses the $\Pr(B)$ convention).

Common random numbers. We use the same random seed for all $(\alpha, \text{treatment})$ pairs, providing a form of common random numbers (CRN) variance reduction. Since the same signal sequence is used for both treatments at each α, treatment-difference estimates have lower variance than if independent seeds were used. Because equilibrium strategies differ across treatments, the CRN benefit is partial—the same signal sequence leads to different selling paths once behaviors diverge. Robustness checks in Section 11 confirm that results are stable across 50 independent seeds.

9 Results

Table 1 reports the equilibrium-predicted average $\pi = \Pr(z = G)$ at the time of each sale across treatments and risk aversion levels, from 10,000 simulated games per cell. Lower values indicate more pessimistic beliefs at sale (investors sell when they believe the bad state is likely).

α	Random 1st seller	Random 2nd seller	Random 3rd seller	Average 1st seller	Average 2nd seller	Average 3rd seller
0	0.188	0.187	0.175	0.188	0.187	0.175
0.1	0.188	0.188	0.187	0.188	0.188	0.188
0.2	0.260	0.198	0.168	0.249	0.193	0.171
0.3	0.295	0.233	0.172	0.291	0.225	0.174
0.4	0.314	0.275	0.200	0.315	0.277	0.203
0.5	0.322	0.302	0.241	0.323	0.305	0.248
0.6	0.324	0.315	0.277	0.325	0.318	0.288
0.7	0.325	0.322	0.303	0.325	0.325	0.321
0.8	0.328	0.324	0.319	0.325	0.325	0.325
0.9	0.359	0.333	0.319	0.331	0.325	0.324

Note: Each cell reports the average $\pi = \Pr(z = G)$ at the time of the kth sale, from 10,000 simulated games using equilibrium strategies. At $\alpha = 0$ (risk neutral), predictions are identical across treatments. The 4th investor never sells in equilibrium ($n = 1$ boundary).

Table 1: Equilibrium predictions: average $\pi = \Pr(z = G)$ at each seller position. The 4th investor never sells ($n = 1$ boundary condition).

9.1 Validation against M&M Table 2

Table 2 of M&M reports the average $\Pr(z = B) = 1 - \pi$ at the first sale for their HIGH information environment ($\mu_B = 0.675$, NoCB). Converting our first-seller π to $\Pr(B)$ for comparison:

	Our π	Our $\Pr(B)$	M&M $\Pr(B)$	Difference
$\alpha = 0$ (risk neutral)	0.188	0.812	0.800	+0.012
$\alpha = 0.5$ (risk averse)	0.322	0.678	0.678	< 0.001

The risk-averse prediction matches exactly. The 1.2pp difference under risk neutrality is a systematic grid-resolution effect, not simulation noise (Section 11 shows the seed-induced standard deviation is < 0.001).

10 ★ Replication diagnostic: M&M Appendix D

★ This section documents our systematic comparison of solver output against five sigma values reported in M&M Appendix D. This goes beyond M&M's published analysis by testing whether the discrepancy is algorithmic, and by implementing four alternative solution methods to isolate the source.

10.1 Target values

M&M Appendix D reports equilibrium mixing probabilities σ at specific (n, π) states for the Random treatment. We compare at five points:

α	n	d	$\Pr(B)$ exact	M&M σ	Our σ	Gap
0.0	4	2	0.8118	0.775	0.801	+0.026
0.0	3	2	0.8118	0.016	0.048	+0.032
0.5	4	1	0.6750	0.516	0.539	+0.023
0.5	3	2	0.8118	0.887	0.911	+0.024
0.5	2	3	0.8996	0.860	0.913	+0.053

Here d is the net bad signal count and $\Pr(B) = 1 - \pi(d)$ from Eq. (5). All five gaps are positive: our solver systematically produces higher σ than M&M.

10.2 Seven solver variants tested

We tested whether the gap could be closed by changing the numerical method:

1. Value iteration, low initialization ($V^{(0)} = 0$) — baseline
2. Value iteration, high initialization ($V^{(0)} = u(v)/(1-\alpha)$)
3. Policy iteration with exact linear system solve (Section 7)
4. Howard improvement with $K = 10$ sweeps
5. Howard improvement with $K = 50$ sweeps
6. Discount convention variant
7. $W = U_{\text{sell}}$ at indifference variant

10.3 Result: all variants identical

All seven methods produce **identical** σ values to four decimal places at all five target points. The RMSE is 0.033 for every method. This rules out:

- **Convergence**: Policy iteration solves V exactly via linear algebra—if value iteration had not converged, the two would differ. They agree to 10^{-6}.
- **Initialization sensitivity**: High and low initial conditions converge to the same fixed point.
- **Insufficient V-sweeps**: Howard improvement with 50 sweeps per σ-update gives the same answer.
- **Discount convention**: Moving λ between H and V gives identical σ (algebraically equivalent).
- **W formula**: Using $W = U_{\text{sell}}$ at indifference gives identical results (mathematically expected since $U_{\text{sell}} = U_{\text{hold}}$ there).

10.4 Interpretation

The systematic 2–5% upward bias in σ is not a numerical or algorithmic artifact. The most likely explanations are:

1. **Grid or interpolation differences.** M&M may use a different grid structure, finer resolution in certain regions, or higher-order interpolation.
2. **Root-finding tolerance.** Different tolerance in Brent's method or a different root-finding algorithm could produce small differences that compound through the value iteration.
3. **Minor errata.** Published appendix values may contain rounding or transcription errors.

Importantly, the sigma-level discrepancy does not propagate to aggregate predictions: our simulated average $\Pr(B)$ at the first sale matches M&M Table 2 within 0.001 for $\alpha = 0.5$ (Section 8.1). The simulation integrates over the full strategy profile and signal process, smoothing out pointwise sigma differences.

11 ★ Robustness checks

★ We test the sensitivity of our results to two key numerical choices: grid resolution ($T_{\max}$) and random seed.

11.1 Grid resolution

We solve the equilibrium at $T_{\max} \in \{10, 20, 40, 80, 160\}$, corresponding to grids of 21, 41, 81, 161, and 321 points.

Sigma convergence. At all five M&M target points, the sigma values are identical (to four decimal places) for $T_{\max} \geq 20$. The largest change from $T_{\max} = 20$ to $T_{\max} = 160$ is below 10^{-3}. Only $T_{\max} = 10$ shows any deviation, with a maximum $|\Delta| = 0.002$.

Threshold convergence. Continuous thresholds $\pi^*(n)$ are even more stable: the maximum change from $T_{\max} = 10$ to $T_{\max} = 160$ is 1.5×10^{-4} (at $\alpha = 0.5$, $n = 2$). For all other $(alpha, n)$ combinations, the change is below 10^{-4}.

Conclusion. The 41-point grid ($T_{\max} = 20$) is more than sufficient for our purposes. The solver output is numerically converged with respect to grid resolution.

11.2 Random seed sensitivity

We run the simulation with 50 independent seeds (seeds 1–50), each producing 10,000 games, for $\alpha \in \{0.0, 0.5\}$ under the Random treatment.

α	Position	Mean Pr(B)	SD	Min	Max
0.0	1st sale	0.812	< 0.001	0.812	0.812
0.0	2nd sale	0.813	0.000	0.813	0.814
0.0	3rd sale	0.825	0.001	0.823	0.826
0.5	1st sale	0.678	0.000	0.678	0.679
0.5	2nd sale	0.698	0.001	0.697	0.700
0.5	3rd sale	0.758	0.001	0.755	0.761

The standard deviation across 50 seeds is at most 0.001 for the 1st and 2nd sales, and 0.001–0.002 for the 3rd sale. The full range (max − min) is at most 0.006. This confirms:

- The 1.2pp gap between our $\alpha = 0$ first-sale prediction (0.812) and M&M's (0.800) is *not* simulation noise—it is approximately 40× the seed-induced standard deviation.
- 10,000 games per cell provides sufficient precision for the comparisons in this paper.
- Later seller positions have modestly higher variance, as expected from smaller sample sizes (not all games reach the 3rd sale).

12 Summary of extensions

The following elements extend beyond the M&M (2020) framework:

1. **Average treatment payoffs** (Eqs. 7, Section 5): treatment-specific ρ and L functions, Jensen's inequality characterization, and the holding-channel-dominates result.

2. **Per-seller belief simulation** (Section 8): tracking average $\Pr(B)$ at each seller position (1st, 2nd, 3rd) with common random numbers across treatments.

3. **Alternative solution algorithms** (Section 7): policy iteration with exact linear system solve, Howard improvement, and convention variants.

4. **Replication diagnostic** (Section 10): systematic comparison of seven solver variants against M&M Appendix D, ruling out algorithmic causes for the 2–5% sigma gap.

5. **Robustness checks** (Section 11): grid-resolution sensitivity ($T_{\max}$ from 10 to 160) and seed sensitivity (50 seeds $\times$ 10,000 games).